CSCE 46103/56103 Artificial Intelligence – Fall 2026

Evaluation, Generalization, Regularization, and Model Selection

Jungtaek Kim
jungtaek.kim@uark.edu

University of Arkansas

October 29, 2026

This Material's Topics

- Evaluating predictions on examples separate from training.
- Understanding generalization and model complexity.
- Controlling model complexity through regularization.
- Selecting hyperparameters with validation and cross-validation.

The Central Question

How do we choose a model that predicts well on future examples?

Model selection and hyperparameter tuning: Russell and Norvig [2020, Secs. 19.4.1 and 19.4.4].

UNIVERSITY OF ARKANSAS

The Restaurant Hypotheses

Recall $x = (\text{Patrons}, \text{WaitEstimate})$ and $y = \text{WillWait}$.

Rule	Prediction	Training errors
h_0	Always predict Yes.	$2/6$
h_1	Predict Yes when Patrons is Some.	$1/6$
h_2	Predict Yes when WaitEstimate is 0–10 or 10–30.	$0/6$

For h_1 and h_2, predict No in all other cases.

Restaurant observations: Russell and Norvig [2020, Fig. 19.2]. Rules and training/validation split are classroom examples.

UNIVERSITY OF ARKANSAS

The Restaurant Hypotheses

Recall $x = (\text{Patrons}, \text{WaitEstimate})$ and $y = \text{WillWait}$.

Rule	Prediction	Training errors
h_0	Always predict Yes.	$2/6$
h_1	Predict Yes when Patrons is Some.	$1/6$
h_2	Predict Yes when WaitEstimate is 0–10 or 10–30.	$0/6$

For h_1 and h_2, predict No in all other cases.

Evaluation on Additional Examples

h_2 makes no mistakes on $x_1, \ldots, x_6$. We next compare the rules on $x_7, \ldots, x_{12}$.

Restaurant observations: Russell and Norvig [2020, Fig. 19.2]. Rules and training/validation split are classroom examples.

Six Validation Examples

Use the remaining restaurant cases as an illustrative validation set.

Example	Patrons	Wait est. (min)	y	$h_1(x)$	$h_2(x)$
x_7	None	0–10	No	No	Yes
x_8	Some	0–10	Yes	Yes	Yes
x_9	Full	> 60	No	No	No
x_{10}	Full	10–30	No	No	Yes
x_{11}	None	0–10	No	No	Yes
x_{12}	Full	30–60	Yes	No	No

Restaurant observations: Russell and Norvig [2020, Fig. 19.2]. Rules and training/validation split are classroom examples.

Six Validation Examples

Use the remaining restaurant cases as an illustrative validation set.

Example	Patrons	Wait est. (min)	y	$h_1(x)$	$h_2(x)$
x_7	None	0–10	No	No	Yes
x_8	Some	0–10	Yes	Yes	Yes
x_9	Full	> 60	No	No	No
x_{10}	Full	10–30	No	No	Yes
x_{11}	None	0–10	No	No	Yes
x_{12}	Full	30–60	Yes	No	No

Counting the Mistakes

h_1 makes one error; h_2 makes four errors.

Restaurant observations: Russell and Norvig [2020, Fig. 19.2]. Rules and training/validation split are classroom examples.

Training and Validation Error

Rule	Training error	Validation error
h_0	$2/6$	$4/6$
h_1	$1/6$	$1/6$
h_2	$0/6$	$4/6$

- Training error measures fit to the examples used for learning.
- Validation error helps compare choices using separate examples.

Restaurant observations: Russell and Norvig [2020, Fig. 19.2]. Rules and training/validation split are classroom examples.

Training and Validation Error

Rule	Training error	Validation error
h_0	$2/6$	$4/6$
h_1	$1/6$	$1/6$
h_2	$0/6$	$4/6$

- Training error measures fit to the examples used for learning.
- Validation error helps compare choices using separate examples.

Model Selection in This Example

Choose h_1 using validation error, even though h_2 has lower training error.

Restaurant observations: Russell and Norvig [2020, Fig. 19.2]. Rules and training/validation split are classroom examples.

Training, Validation, and Test Sets

Subset	Purpose	Example use
Training	Fit the model's parameters.	Learn the coefficients of a polynomial.
Validation	Compare model choices.	Select its degree.
Test	Evaluate the selected procedure.	Report its error on fresh examples.

Separate Roles

Keep the test set aside while choosing the model and its settings.

Model selection and hyperparameter tuning: Russell and Norvig [2020, Secs. 19.4.1 and 19.4.4].

Average Loss on an Evaluation Set

For an evaluation set S of m labeled examples,

$$\widehat{R}_S(h) = \frac{1}{m} \sum_{(x,y)\in S} L(h(x), y).$$

Classification Error

Use zero–one loss:

$$L(\hat{y}, y) = \mathbf{1}[\hat{y} \neq y].$$

Average loss is the fraction of incorrect predictions.

Mean Squared Error

Use squared loss:

$$L(\hat{y}, y) = (\hat{y} - y)^2.$$

Root mean squared error, $\mathrm{RMSE} = \sqrt{\mathrm{MSE}}$, has the target's units.

Evaluation and loss: Russell and Norvig [2020, Sec. 19.4].

A Confusion Matrix

For h_2 on the six validation cases, take Yes as the positive class.

	Actual Yes	**Actual No**
Predicted Yes	True positive: 1	False positive: 3
Predicted No	False negative: 1	True negative: 1

- A false positive predicts that a customer will wait, but they do not.
- A false negative predicts that a customer will not wait, but they do.

Restaurant observations: Russell and Norvig [2020, Fig. 19.2]. Rules and training/validation split are classroom examples.

Accuracy, Precision, and Recall

Metric	Question	For h_2
Accuracy	What fraction of all predictions are correct?	$\frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} = \frac{1}{3}$
Precision	Of the predicted positives, what fraction are positive?	$\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} = \frac{1}{4}$
Recall	Of the actual positives, what fraction are found?	$\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} = \frac{1}{2}$

Checkpoint

For h_1, $\mathrm{TP} = 1$, $\mathrm{FP} = 0$, $\mathrm{FN} = 1$, $\mathrm{TN} = 4$. What are its precision and recall?

Restaurant observations: Russell and Norvig [2020, Fig. 19.2]. Rules and training/validation split are classroom examples.

UNIVERSITY OF ARKANSAS

Accuracy, Precision, and Recall

Metric	Question	For h_2
Accuracy	What fraction of all predictions are correct?	$\frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} = \frac{1}{3}$
Precision	Of the predicted positives, what fraction are positive?	$\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} = \frac{1}{4}$
Recall	Of the actual positives, what fraction are found?	$\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} = \frac{1}{2}$

Checkpoint

For h_1, $\mathrm{TP} = 1$, $\mathrm{FP} = 0$, $\mathrm{FN} = 1$, $\mathrm{TN} = 4$. What are its precision and recall?
Precision is 1; recall is $1/2$.

Restaurant observations: Russell and Norvig [2020, Fig. 19.2]. Rules and training/validation split are classroom examples.

Loss and Task Goals

A loss function assigns a cost to a prediction error.

- Zero–one loss treats all classification mistakes equally.
- Different error types can have different costs in the application.
- Squared loss gives large numerical errors more weight than small ones.

An Unequal-Cost Classification Loss

$$L(\hat{y}, y) = \begin{cases} 0, & \hat{y} = y, \\ c_{\mathrm{FP}}, & \hat{y} = \text{Yes},\ y = \text{No}, \\ c_{\mathrm{FN}}, & \hat{y} = \text{No},\ y = \text{Yes}. \end{cases}$$

Choose the costs to represent the consequences of the errors.

Evaluation and loss: Russell and Norvig [2020, Sec. 19.4].

Generalization

Generalization is a learned model's performance on new examples.

Empirical Risk

$$\widehat{R}_S(h) = \frac{1}{|S|} \sum_{(x,y)\in S} L(h(x), y).$$

Average loss on a finite set of examples.

Population Risk

$$R(h) = \mathbb{E}_{(X,Y)\sim P}[L(h(X), Y)].$$

Expected loss for the distribution P of future examples.

A separate evaluation sample from P lets us estimate a fixed model's population risk.

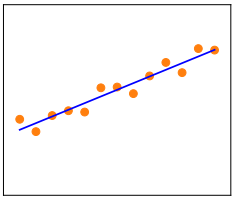
Evaluation and loss: Russell and Norvig [2020, Sec. 19.4].

UNIVERSITY OF ARKANSAS

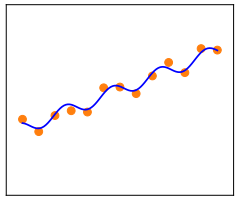
Hypothesis Spaces

Different hypothesis spaces fit the same sample in different ways.

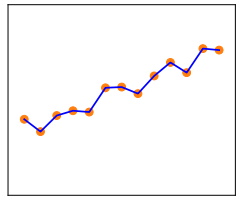
Linear



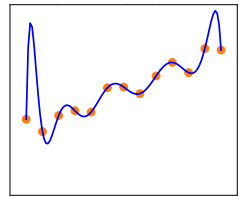
Sinusoidal



Piecewise linear



Degree-12 polynomial



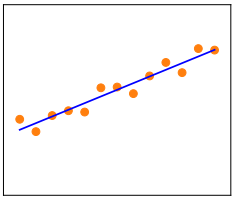
Horizontal axis: input x. Vertical axis: target y.

Textbook fits from Russell and Norvig [2020, Fig. 19.1, top row].

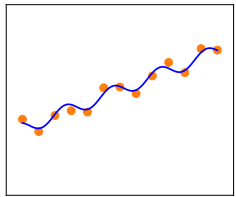
Hypothesis Spaces

Different hypothesis spaces fit the same sample in different ways.

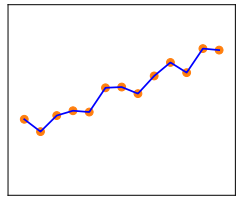
Linear



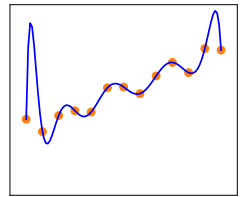
Sinusoidal



Piecewise linear



Degree-12 polynomial

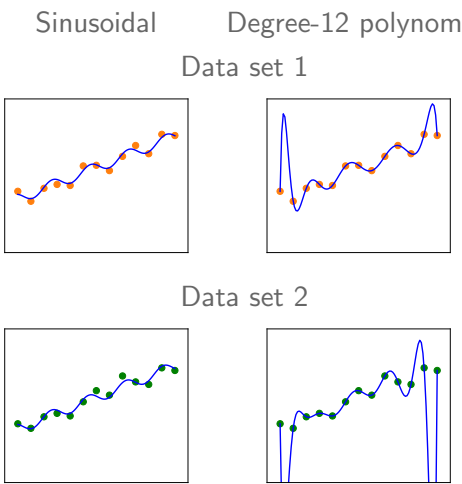


Horizontal axis: input x. Vertical axis: target y.

- A straight line captures the overall trend.
- More flexible functions can follow finer variations in the sample.
- Our goal is to capture structure that continues beyond these observations.

Textbook fits from Russell and Norvig [2020, Fig. 19.1, top row].

Sensitivity to the Training Sample



- Both samples come from the same underlying function.
- The sinusoidal fit keeps a similar shape.
- The high-degree polynomial changes sharply between observations.

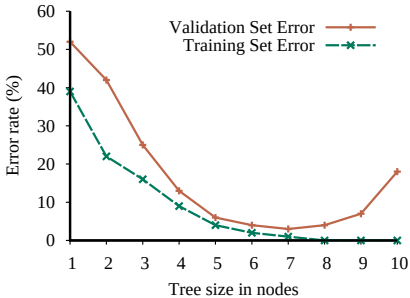
Variance

A model has high variance when its predictions change substantially across training samples.

Selected panels from Russell and Norvig [2020, Fig. 19.1]. Axes: input x and target y.

Model Complexity and Prediction Error

A decision tree tests features at internal nodes and predicts at its leaves.



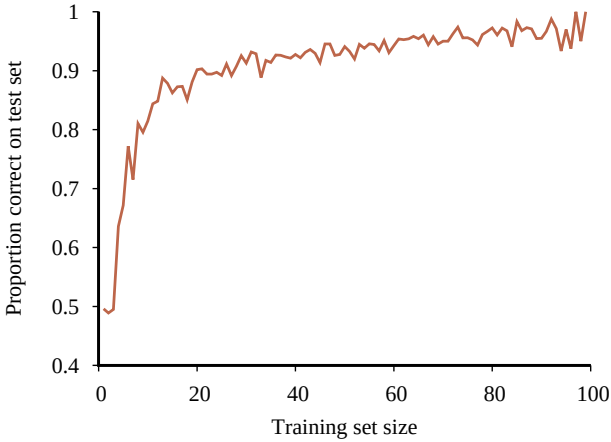
Choose the model size using validation error.

- Small trees underfit: both errors are high.
- Training error falls as tree size grows.
- Validation error is lowest at seven nodes.
- Larger trees overfit in this example.

Restaurant decision-tree example from Russell and Norvig [2020, Fig. 19.9(a)].

UNIVERSITY OF ARKANSAS

Learning Curves



- Horizontal axis: number of training examples.
- Vertical axis: proportion of test predictions that are correct.
- More examples improve the learned restaurant decision tree.

A Learning Curve

Tracks performance as training-set size changes.

Russell and Norvig [2020, Fig. 19.7]: restaurant-domain examples; each point averages 20 trials.

Discussion: Expected Proportion Correct

Average accuracy over i.i.d. training sets of size n and fresh test examples.

Must expected proportion correct increase whenever n increases?

Discussion: Expected Proportion Correct

Average accuracy over i.i.d. training sets of size n and fresh test examples.

Must expected proportion correct increase whenever n increases?

No informative features; $P(Y = 1) = 0.6$ and $P(Y = 0) = 0.4$.
Predict the training-label majority on every example; break ties toward 0.

Discussion: Expected Proportion Correct

Average accuracy over i.i.d. training sets of size n and fresh test examples.

Must expected proportion correct increase whenever n increases?

No informative features; $P(Y = 1) = 0.6$ and $P(Y = 0) = 0.4$.
Predict the training-label majority on every example; break ties toward 0.

One Training Example

Predict 1 with probability 0.6.

$$0.6(0.6) + 0.4(0.4) = 0.52.$$

Two Training Examples

Predict 1 only if both labels are 1:
probability $0.6^2 = 0.36$.

$$0.36(0.6) + 0.64(0.4) = 0.472.$$

Improvement Need Not Be Monotonic

Expected accuracy drops from 52% to 47.2%.
It approaches 60% as $n \to \infty$, but need not improve at every step.

Checkpoint: Error Patterns

Two models are evaluated with the same loss and data split.

	Training error	Validation error
Model A	High	High
Model B	Low	Much higher than training

Which pattern suggests underfitting? Which suggests overfitting?

Evaluation and loss: Russell and Norvig [2020, Sec. 19.4].

Checkpoint: Error Patterns

Two models are evaluated with the same loss and data split.

	Training error	Validation error
Model A	High	High
Model B	Low	Much higher than training

Which pattern suggests underfitting? Which suggests overfitting?

Interpretation

A suggests underfitting: try a more expressive model or better features.
B suggests overfitting: try regularization, a simpler model, or more training data.

Evaluation and loss: Russell and Norvig [2020, Sec. 19.4].

Regularized Empirical Risk

Regularization adds a preference for simpler hypotheses to the objective:

$$\hat{h}_\lambda = \arg\min_{h \in \mathcal{H}} \left[\widehat{R}_{D_{\text{train}}}(h) + \lambda \Omega(h) \right].$$

- $\widehat{R}_{D_{\text{train}}}(h)$ measures fit to the training data.
- $\Omega(h)$ penalizes the chosen notion of complexity.
- $\lambda \geq 0$ controls the strength of the penalty.

Regularization: Russell and Norvig [2020, Sec. 19.4.3].

Regularized Empirical Risk

Regularization adds a preference for simpler hypotheses to the objective:

$$\hat{h}_\lambda = \arg\min_{h \in \mathcal{H}} \left[\widehat{R}_{D_{\text{train}}}(h) + \lambda \Omega(h) \right].$$

- $\widehat{R}_{D_{\text{train}}}(h)$ measures fit to the training data.
- $\Omega(h)$ penalizes the chosen notion of complexity.
- $\lambda \geq 0$ controls the strength of the penalty.

The Tradeoff

Accept some additional training loss when it gives a model that generalizes better.

Regularization: Russell and Norvig [2020, Sec. 19.4.3].

L2 Regularization

For a linear predictor $h_{\mathbf{w},b}(x) = \mathbf{w}^\mathsf{T} x + b$, let $\mathbf{w}$ contain the d feature weights and b be the intercept. The L2 penalty is

$$\Omega(\mathbf{w}) = \|\mathbf{w}\|_2^2 = \sum_{j=1}^{d} w_j^2.$$

With squared loss, the training objective becomes

$$\frac{1}{n} \sum_{i=1}^{n} \left(\mathbf{w}^\mathsf{T} x_i + b - y_i \right)^2 + \lambda \sum_{j=1}^{d} w_j^2.$$

- Large coefficients receive a larger penalty.
- Increasing λ favors smaller coefficient magnitudes.
- The intercept b is commonly left unpenalized.

Regularization: Russell and Norvig [2020, Sec. 19.4.3].

A One-Weight Example

Suppose the training loss is $(w-2)^2$. Add an L2 penalty:

$$J_\lambda(w) = (w-2)^2 + \lambda w^2.$$

A One-Weight Example

Suppose the training loss is $(w-2)^2$. Add an L2 penalty:

$$J_\lambda(w) = (w-2)^2 + \lambda w^2.$$

$$\frac{\mathrm{d}J_\lambda}{\mathrm{d}w} = 2(w-2) + 2\lambda w = 0 \quad \Longrightarrow \quad w_\lambda^* = \frac{2}{1+\lambda}.$$

λ	**Optimal weight** w_λ^*
0	2
1	1
3	$\frac{1}{2}$

A stronger penalty moves the fitted weight toward zero.

Regularization: Russell and Norvig [2020, Sec. 19.4.3].

The Regularization Strength

Compare two candidate models using $J_\lambda(h) = \widehat{R}(h) + \lambda\Omega(h)$.

	Training loss $\widehat{R}$	**Penalty Ω**
A	0.10	9
B	0.25	1

Regularization: Russell and Norvig [2020, Sec. 19.4.3].

The Regularization Strength

Compare two candidate models using $J_\lambda(h) = \widehat{R}(h) + \lambda\Omega(h)$.

	Training loss $\widehat{R}$	**Penalty Ω**
A	0.10	9
B	0.25	1

λ	$J_\lambda(A)$	$J_\lambda(B)$	**Selected model**
0.01	0.19	0.26	A
0.10	1.00	0.35	B

Use validation performance to choose λ; then train with that value.

Regularization: Russell and Norvig [2020, Sec. 19.4.3].

L1 Regularization

An L1 penalty uses absolute coefficient values:

$$\Omega(\mathbf{w}) = \|\mathbf{w}\|_1 = \sum_{j=1}^{d} |w_j|.$$

- L1 regularization can set some coefficients exactly to zero.
- A model with many zero coefficients is called sparse.
- A zero coefficient removes that feature's contribution to a linear predictor.

Example of Sparsity

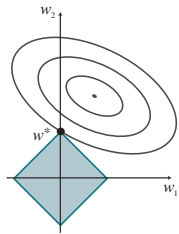
$h(x) = 1.2x_1 + 0x_2 - 0.4x_3 + b$ uses x_1 and x_3 to make predictions.

Regularization: Russell and Norvig [2020, Sec. 19.4.3].

L1 and L2 Geometry

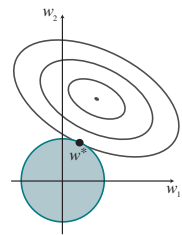
A weight budget t limits the shaded region. Expand equal-loss contours until they first touch it at $\mathbf{w}^*$. Smaller t restricts the weights more.

L1 constraint: $|w_1| + |w_2| \leq t$



The diamond has corners on the axes, where a coefficient is zero.

L2 constraint: $w_1^2 + w_2^2 \leq t$



The circle usually meets a contour away from the axes.

Constrained view of regularization; Russell and Norvig [2020, Fig. 19.14].

Discussion: L1 and Sparsity

For convex, differentiable loss $\ell(w)$ and $\lambda \geq 0$, let $J(w) = \ell(w) + \lambda|w|$.

When does $w = 0$ minimize J?

One-Sided Slopes at Zero

$$J'_-(0) = \ell'(0) - \lambda,$$
$$J'_+(0) = \ell'(0) + \lambda.$$

$w = 0$ minimizes J exactly when

$$J'_-(0) \leq 0 \leq J'_+(0)$$
$$\iff \quad \boxed{|\ell'(0)| \leq \lambda}.$$

The One-Weight Example

$\ell(w) = (w-2)^2$, so $\ell'(0) = -4$.
Zero is optimal exactly when $\lambda \geq 4$.

λ	$J'_-(0)$	$J'_+(0)$	w^*
3	-7	-1	0.5
4	-8	0	0
5	-9	1	0

A range of loss slopes makes zero optimal. With other weights fixed, this condition can hold for many coefficients, producing sparsity.

Checkpoint: Regularization

A polynomial regressor fits the training data closely but has high validation error.

1. What could happen if we increase its L2 penalty?
2. What could happen if the penalty becomes too strong?
3. Which data should guide the choice of penalty strength?

Regularization: Russell and Norvig [2020, Sec. 19.4.3].

Checkpoint: Regularization

A polynomial regressor fits the training data closely but has high validation error.

1. What could happen if we increase its L2 penalty?
2. What could happen if the penalty becomes too strong?
3. Which data should guide the choice of penalty strength?

Answers

A larger penalty favors smaller coefficients and can reduce overfitting.
An excessive penalty can cause underfitting.
Choose the strength using validation performance.

Regularization: Russell and Norvig [2020, Sec. 19.4.3].

Parameters and Hyperparameters

Model	Parameters learned from training	Hyperparameters selected by validation
Linear regression	Coefficients and intercept	Penalty type and strength
Polynomial regression	Polynomial coefficients	Maximum degree
Decision tree	Split tests and leaf predictions	Maximum depth or size limit

Two Levels of Choice

Fit parameters for each setting; compare the resulting models to choose the setting.

Model selection and hyperparameter tuning: Russell and Norvig [2020, Secs. 19.4.1 and 19.4.4].

Validation-Based Model Selection

For candidate hyperparameter settings $\alpha \in \mathcal{A}$:

1. Train a model $\hat{h}_\alpha$ on D_{train}.
2. Measure its loss on D_{val}.
3. Choose the setting with the smallest validation loss:

$$\alpha^* = \arg\min_{\alpha \in \mathcal{A}} \widehat{R}_{D_{\text{val}}}(\hat{h}_\alpha).$$

Model selection and hyperparameter tuning: Russell and Norvig [2020, Secs. 19.4.1 and 19.4.4].

Validation-Based Model Selection

For candidate hyperparameter settings $\alpha \in \mathcal{A}$:

1. Train a model $\hat{h}_\alpha$ on D_{train}.
2. Measure its loss on D_{val}.
3. Choose the setting with the smallest validation loss:

$$\alpha^* = \arg\min_{\alpha \in \mathcal{A}} \widehat{R}_{D_{\text{val}}}(\hat{h}_\alpha).$$

Example: A Grid of Settings

Try polynomial degree $d \in \{1, 2, 3, 4\}$ and L2 strength $\lambda \in \{0, 0.01, 0.1, 1\}$. This gives $4 \times 4 = 16$ combinations.

Model selection and hyperparameter tuning: Russell and Norvig [2020, Secs. 19.4.1 and 19.4.4].

K-Fold Cross-Validation

Reserve the test set. Partition the remaining development data into K folds.

Round	Validation fold	Training folds
1	1	2, 3, 4, 5
2	2	1, 3, 4, 5
3	3	1, 2, 4, 5
4	4	1, 2, 3, 5
5	5	1, 2, 3, 4

In each round, fit preprocessing using only the training folds, then apply it to both training and validation examples.

For each candidate setting, train a fresh model in each round and average its validation losses.

Model selection and hyperparameter tuning: Russell and Norvig [2020, Secs. 19.4.1 and 19.4.4].

A Cross-Validation Example

Compare maximum tree depths using five equal-sized folds.

Max. depth	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean error
1	0.30	0.25	0.35	0.30	0.30	0.30
2	0.20	0.15	0.20	0.25	0.20	0.20
3	0.25	0.20	0.30	0.20	0.30	0.25

Illustrative fold errors for a worked exercise. Cross-validation: Russell and Norvig [2020, Sec. 19.4.1].

A Cross-Validation Example

Compare maximum tree depths using five equal-sized folds.

Max. depth	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean error
1	0.30	0.25	0.35	0.30	0.30	0.30
2	0.20	0.15	0.20	0.25	0.20	0.20
3	0.25	0.20	0.30	0.20	0.30	0.25

The Selected Setting

Choose maximum depth 2. Cross-validation compares settings, each evaluated through several fitted models.

Illustrative fold errors for a worked exercise. Cross-validation: Russell and Norvig [2020, Sec. 19.4.1].

Choosing a Data Split

The split should reflect the predictions we want the model to make.

Data setting	Splitting approach	Purpose
Independent examples	Random split; stratify classification labels when useful	Keep class proportions similar.
Several records per person or episode	Keep a whole group in one subset	Evaluate on new people or episodes.
Time-ordered observations	Train on the past; evaluate on later periods	Evaluate future forecasting.

Developing machine-learning systems: Russell and Norvig [2020, Sec. 19.9].

UNIVERSITY OF ARKANSAS

Refitting and Final Evaluation

After selecting the hyperparameters:

1. Fix the chosen model family, preprocessing steps, and settings.
2. Refit the full procedure on all development data.
3. Evaluate the resulting model on the held-out test set.
4. Report the selected metric and compare with a simple baseline.

Using the Available Data

Development data include the training and validation examples. After selection, they can all contribute to the final fit.

Model selection and hyperparameter tuning: Russell and Norvig [2020, Secs. 19.4.1 and 19.4.4].

Restaurant Model Selection

1. Define the task: predict WillWait from restaurant attributes.
2. Set aside test examples and choose classification error as the metric.
3. Compare tree-depth limits by cross-validation on the development data.
4. Select maximum depth 2 in our worked cross-validation example.
5. Learn a new tree on all development examples with that depth limit.
6. Evaluate its predictions on the test examples.

Restaurant task: Russell and Norvig [2020, Sec. 19.3]. Model-selection procedure: Sec. 19.4; depth choice refers to the classroom exercise.

Restaurant Model Selection

1. Define the task: predict WillWait from restaurant attributes.
2. Set aside test examples and choose classification error as the metric.
3. Compare tree-depth limits by cross-validation on the development data.
4. Select maximum depth 2 in our worked cross-validation example.
5. Learn a new tree on all development examples with that depth limit.
6. Evaluate its predictions on the test examples.

The final tree learns its split tests and leaf predictions during refitting.

Restaurant task: Russell and Norvig [2020, Sec. 19.3]. Model-selection procedure: Sec. 19.4; depth choice refers to the classroom exercise.

Checkpoint: Model Selection

Three regularization strengths produce these errors:

λ	**Training error**	**Validation error**
0	0.05	0.18
1	0.08	0.12
10	0.15	0.16

Which setting should we select? What do we do next?

Model selection and hyperparameter tuning: Russell and Norvig [2020, Secs. 19.4.1 and 19.4.4].

Checkpoint: Model Selection

Three regularization strengths produce these errors:

λ	**Training error**	**Validation error**
0	0.05	0.18
1	0.08	0.12
10	0.15	0.16

Which setting should we select? What do we do next?

Answer

Select $\lambda = 1$, refit on the development data, and evaluate on the test set. Validation error determines the choice.

Model selection and hyperparameter tuning: Russell and Norvig [2020, Secs. 19.4.1 and 19.4.4].

Main Takeaways

- Evaluation compares predictions with targets using a task-appropriate loss or metric.
- Generalization concerns performance on new examples.
- Model complexity affects both fit and sensitivity to training data.
- Regularization balances training fit with a preference for simpler models.
- Validation and cross-validation guide model selection; a held-out test set evaluates the final result.

Model selection and hyperparameter tuning: Russell and Norvig [2020, Secs. 19.4.1 and 19.4.4].

Thank You

Any questions?

References

S. J. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, 4th edition, 2020. URL `https://aima.cs.berkeley.edu/`.